

# Module 7: Introduction to Snowflake

## Learning Objectives

Understand the fundamentals of Snowflake and its core components used in modern cloud data engineering.

### 1. Cloud Data Warehouse

Snowflake is a cloud-native data warehouse that stores and analyzes structured and semi-structured data. It separates storage and compute, enabling scalable and cost-effective analytics.

### 2. Snowflake Architecture

Snowflake consists of three layers:

- Database Storage Layer
- Compute Layer (Virtual Warehouses)
- Cloud Services Layer (Security, Metadata, Query Optimization).

### 3. Database

A database is the top-level logical container that stores schemas and objects.

### 4. Schema

A schema organizes related database objects such as tables, views, and stages.

### 5. Tables

Tables store business data. Snowflake supports Permanent, Temporary, and Transient tables.

### 6. Virtual Warehouse

A Virtual Warehouse is the compute engine that executes SQL queries independently from storage.

### 7. Stages

Stages are locations used to load and unload files such as CSV, JSON, and Parquet into Snowflake. Internal and External stages are supported.

## Key Takeaways

- Fully managed cloud platform
- Independent scaling of storage and compute
- Secure and high-performance analytics
- Simple data loading using stages
- Supports SQL, Python, AI, and Data Engineering workloads